



Thanasak Wanglomklang

Mechanical Engineer / AI Researcher

CONTACT

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EDUCATION

PhD · Mechanical Engineering
École Centrale de Lyon, France
2023 – Present

M.Eng · Mechatronic Engineering
Suranaree Univ. of Technology, TH
2018 – 2019
GPAX 4.00 / 4.00

B.Eng · Mechanical Engineering
Suranaree Univ. of Technology, TH
2014 – 2018 First Class Honors
GPAX 3.94 / 4.00

LANGUAGES

Thai	Native
English	Professional Working
French	Professional Working
Japanese	Elementary

PROGRAMMING

Python
MATLAB / Simulink
C / C++

COMPUTATIONAL TOOLS

Software
COMSOL Multiphysics, MATLAB,
SolidWorks, Gmsh, ParaView,
PyVista, sensors / DAQ

Python Libraries
Scientific NumPy, SciPy, scikit-learn
Deep Learning PyTorch, GPyTorch
Optimization pymoo, SALib, PyEGRO
Visualization Matplotlib, PyVista
Meshing Gmsh API, mph (COMSOL)

HPC / Dev

SLURM, Linux, Git,
PySide6/Tkinter (GUI)

PROFESSIONAL PROFILE

Mechanical engineer and Physics-AI researcher connecting **system dynamics, measured signals, and computational modeling** to solve practical engineering problems. Background spans precision automation, vibration diagnostics, physics-based simulation, and uncertainty-aware optimization across manufacturing, acoustics, aerodynamics, and materials design. Develops interpretable workflows and software tools that turn sensor data and high-fidelity simulations into design insight, diagnostic decisions, and validated research outcomes.

Expertise: Control systems design, physics-informed AI, surrogate modeling, autoencoder-based reduced-order modeling, sensitivity and uncertainty analysis, vibration diagnostics, system identification, MATLAB/Simulink, Python/C/C++, and engineering software development.

PROFESSIONAL EXPERIENCE

JSPS Short-Term Postdoctoral Fellow

Apr 2026 – Present

Kyushu University · Fukuoka, Japan

► Developing physics-informed autoencoder/GNN workflows for nonlinear 3D aerodynamic shape reduction and data-efficient surrogate optimization.

Visiting Scientist

May – Jun 2025

AIMR, Tohoku University · Sendai, Japan

► Developed a Physics-AI inverse-design workflow for microscale woven material cells using reduced representations and FEM validation.

Lecturer - Mechanical & Mechatronics

Dec 2019 – Sep 2023

Suranaree University of Technology · Thailand

- Taught control systems and structural vibration at undergraduate and graduate levels.
- Supervised student projects in mechatronics, fault diagnostics, and system modeling.
- Co-authored peer-reviewed publications; delivered industrial training in vibration analysis.

R&D Master Intern - Precision Automation Controls

Mar 2018 – Oct 2019

Western Digital Storage Technologies (Thailand) Ltd.

► Built data-driven diagnostics from vibration, motor-current, encoder/linear-bearing behavior, and image-derived features for HGA precision-assembly equipment.
► Connected machine-condition signals with positioning behavior and process reliability for monitoring, calibration, and fault detection.

SELECTED ENGINEERING PROJECTS

Observer-Based PI Servo Control for HGA Automation

2018 – 2020

Western Digital Storage Technologies (Thailand) Ltd. · ACAM / Slider Attach

Modeled feed-drive actuator dynamics and linked PI/observer response, encoder/linear-axis faults, positioning errors, and adhesive-dispense indicators to support calibration, monitoring, and slider-attach reliability.

Data Acquisition & FDC for Precision Assembly Tools

2018 – 2020

Western Digital Storage Technologies (Thailand) Ltd. · Sensors / Vibration / Vision Features

Built DAQ/FDC workflows using vibration, motor current, and image-derived features with ANOVA, spectrum analysis, and ANN classification to diagnose mount-head, vacuum, bearing, and slider-loss conditions before process alarms.

AI-Driven Inverse Design of Woven Material Cells

2024 – 2025

AIMR Tohoku Univ., Japan / LTDS ECL, France · Collaborative

Developed a semi-supervised ConvAE workflow that reduced a large woven-pattern design space into a compact latent manifold for surrogate-based topology optimization, with FEM-validated errors below 5%.

PyEGRO: Open-Source Optimization Under Uncertainty

2024

Individual Contributor · Open Source · twanglom.github.io/PyEGRO

Open-source Python library for engineering robust design under uncertainty, integrating GPR/Co-Kriging surrogates, adaptive sampling, and multi-objective evolutionary search.

MechIntel Digital Twin Platform

2026 – Present

Individual Contributor · 3D Assets / Sensors / Simulation / Diagnostics

► Building a Digital Twin web platform connecting physical assets, CAD/3D models, simulation outputs, and operational data into live engineering twins.
► Integrating OpenUSD/GLB visualization workflows with sensor/PLC data for monitoring, diagnostics, and predictive-maintenance use cases.

CORE COMPETENCIES

AI & Deep Learning	Physics-informed AI, autoencoder-based reduced-order modeling, Gaussian Process Regression, Co-Kriging, surrogate modeling, PyTorch, GPyTorch, scikit-learn.
Optimization & UQ	Design of Experiments (DOE), surrogate-based optimization, robust design optimization, NSGA-II/pymoo, Sobol sensitivity analysis, adaptive sampling, uncertainty propagation.
Vibration & Dynamics	Structural vibration, system dynamics, frequency-domain analysis, actuator/bearing behavior, sensors/DAQ, metrology, vibration-based condition monitoring.
Control Systems	PI/PID control, observer-based diagnostics, state feedback, closed-loop control, system identification, digital control, MATLAB/Simulink.
Scientific Software	Python/C/C++, COMSOL, SolidWorks, Gmsh, ParaView, PyVista, Git, PySide6/Tkinter GUI, open-source engineering tools.

AWARDS & FELLOWSHIPS

2026–27	JSPS Short-Term Fellowship	Japan Society for the Promotion of Science (JSPS), Kyushu University. Competitive fellowship for international researchers; selected for AI-driven engineering design research.
2023–26	Franco-Thai Scholarship	Campus France / France Government: PhD scholarship for outstanding Thai researchers in France.
2024–25	Research & Mobility Grants	MES-FR / ELyTMaX / CNRS/ MEGA Doctoral School/ École Centrale de Lyon. Grants supporting research visits to Tohoku University (2025) and Kyushu University (2026).
2019	Best Presentation Award	MERE 2019 International Conference, Wuhan, China.
2019	Master’s Scholarship	Western Digital (Thailand). Industry scholarship for M.Eng in precision control and fault diagnostics.

SELECTED PUBLICATIONS

Full list: Google Scholar

[J1] Wanglomklang, T., Shimoyama, K., Gillot, F., & Besset, S. (2026). Robust shape optimization of acoustic diffusers for spatial sound distribution using Co-Kriging metamodel. *Applied Acoustics*, 242, 111047. doi:10.1016/j.apacoust.2025.111047.

[J2] Wanglomklang, T., Shimoyama, K., Gillot, F., & Besset, S. (2025). Neural-network-based two-stage robust shape optimization for acoustic noise reduction in an aircraft cabin. *Engineering Optimization*, 1–19. doi:10.1080/0305215X.2025.2563669.

[J3] Wanglomklang, T., Gillot, F., & Besset, S. (2025). Hybrid Method for Energy flow in Mid-High Frequency Acoustics: Applications in Robust Shape Optimization for Complex Cavities. *Journal of Theoretical and Computational Acoustics*. doi:10.1142/S2591728525500033.

[J4] Chommuangpuck, P., Wanglomklang, T., Tantrairatn, S., & Srisertpol, J. (2020). Fault tolerant control based on an observer on PI servo design for a high-speed automation machine. *Machines*, 8(2), 22.

[J5] Wanglomklang, T., Chommaungpuck, P., & Srisertpol, J. (2020). Linear bearing fault detection using an ANN based on a PI servo system with observer for high-speed automation machine. *IOP Conf. Series: Materials Science and Engineering*, 717(1), 012011.

[J6] Wanglomklang, T., Chommaungpuck, P., Chamniprasart, K., & Srisertpol, J. (2022). Using fault detection and classification techniques for machine breakdown reduction of the HGA process caused by the slider loss defect. *Manufacturing Review*, 9, 21.

[J7] Wanglomklang, T., Tuntavesesak, T., Tumthong, W., & Srisertpol, J. (2022). Roller bearing faults classification using ANN based on servo system with observer design. *WSEAS Transactions on Systems*, 21, 241–246.

INTERNATIONAL CONFERENCES

2026	ELyT Global Workshop	Sendai, Japan (ELyTMaX / CNRS / Tohoku University)
2025	ICFD 2025	Sendai, Japan
2025	WCSMO 2025	Kobe, Japan
2025	DTE/AICOMAS 2025	Paris, France
2024	ECCOMAS Congress 2024	Lisboa, Portugal
2022	ELECS 2022	Bern, Switzerland
2019	MERE 2019	Wuhan, China
2019	E-NETT 2019	Thailand

RESEARCH NETWORKS & COLLABORATORS

LTDS, École Centrale de Lyon — Assoc. Prof. S. Besset; Assoc. Prof. F. Gillot

AIMR, Tohoku University — Assist. Prof. S. Mahmoudi · IRL ELyTMaX / IRN ELyT Global joint lab

Fluids Engineering Design Lab, Kyushu University — Prof. K. Shimoyama